The Science Behind the Optoelectronic Time Code Latch

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1. Purpose and scope

This paper explains the physics and electronics behind the optoelectronic circuit described in the walkthrough: an infrared laser on an optical bench, a reflector that splits light between two detectors, two parallel drivers (A and B), an IF/THEN link built from a capacitor and an NPN transistor, OR/AND gating, a time code Y, and a 74HC74 D flip-flop whose output is called SELECT. Each stage rests on well-established science, which is described below, together with what the complete system can and cannot demonstrate.

Reading of the circuit. The description below follows the supplied schematic and annotations. Component values (resistances, capacitances, supply voltage) were not given, so any numbers shown are illustrative.

2. System overview

Stage	Element	Underlying science
Source	IR laser	Stimulated emission; coherent, near-monochromatic light
Splitting	Reflector (optical bench)	Partial reflection and transmission; intensity division
Sensing	IR detectors A and B	Photoelectric effect in a semiconductor junction
Drive	Driver A (randomized), Driver B (deterministic)	Signal conditioning; noise versus fixed mapping from input x
Threshold	IF/THEN link (C + NPN)	RC charging; base-emitter turn-on near 0.7 V
Logic	OR, AND	Boolean algebra implemented in CMOS or TTL
Timing	Clock / timer, 10 Hz (T = 100 ms)	Periodic sampling
Memory	74HC74 D flip-flop; time code latch	Edge-triggered sequential logic

3. The optical bench: light, splitting and detection

An infrared laser emits light by stimulated emission, in which a photon induces an excited atom or carrier to emit a second photon with the same phase, direction and wavelength. The result is a narrow, stable beam. A reflector placed in the path divides the beam into two portions. If the reflectivity is R and the transmissivity is T, then (neglecting loss) R + T = 1, and the powers reaching the two detectors are $P_A = R\,P_0$ and $P_B = T\,P_0$.

Each detector is a photodiode. A photon with energy above the semiconductor band gap creates an electron-hole pair (the photoelectric effect), and the resulting photocurrent is, to a good approximation, proportional to incident optical power:

$$I_{ph} = (R_{esp}) \times P, \text{ with responsivity } R_{esp} = \eta e \lambda / (h c)$$

where eta is quantum efficiency, e the electron charge, lambda the wavelength, h Planck's constant and c the speed of light. A change in beam intensity (the 'delta intensity' in the schematic) therefore appears as a proportional change in the currents at detectors A and B, which is what lets the optical stage act as a signal source for the electronics.

Where randomness can come from. Even a perfectly steady laser delivers photons at random times. Photon counts follow Poisson statistics: for a mean count N, the standard deviation is the square root of N, so the relative fluctuation is 1/sqrt(N). This *shot noise*, together with thermal (Johnson) noise in the electronics and laser intensity noise, is the physical entropy available to a 'randomized' driver. Raw noise is not uniformly distributed bits, so practical random number generators condition it (for example by thresholding and debiasing). A laser-and-photodiode source of this kind is a physical noise source; it is not a single-photon quantum random number generator.

4. Drivers A and B: random versus deterministic

Driver B is labelled deterministic and takes the problem instance x as input. Its output is a fixed function of x: the same x always produces the same drive signal. Driver A is labelled randomized: its output depends on x and also on noise, so repeated runs on the same x can differ. Running both in parallel produces two signals, A and B, one with a stochastic component and one without. Comparing them is a standard way to separate the effect of the input from the effect of noise.

5. The IF/THEN link: RC integration and a transistor threshold

The IF/THEN link is a threshold circuit. A capacitor C charges through a resistance R toward a supply voltage V_s. Its voltage follows the exponential law

$$V(t) = V_s \ (1 - e^{-t/RC})$$

The base of an NPN silicon bipolar transistor is connected to this voltage. The transistor stays essentially off until the base-emitter voltage reaches about 0.6 to 0.7 V, after which collector current rises exponentially with base voltage (the Shockley diode relation). The circuit therefore implements 'IF V(C) > 0.7 V THEN output HIGH'. Because C needs time to charge, the link also integrates its input over time: a brief pulse may not reach threshold, whereas a sustained signal will. Solving the charging law for the threshold time gives

$$t_{th} = -RC \ \ln(1 - V_{th}/V_s)$$

As an illustration, with R = 100 kilohms, C = 1 microfarad, V_s = 5 V and V_{th} = 0.7 V, RC = 0.1 s and t_{th} is about 15 ms. This delay is the 'memory' that lets the link respond to how long a condition lasted, not merely whether it occurred.

6. OR / AND gating

The outputs A and B (and the IF/THEN output) feed an OR gate and an AND gate. In Boolean terms OR is true when at least one input is true, and AND is true only when all inputs are true. Together they give two different summaries of the same pair of signals:

A	B	A OR B	A AND B
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1

OR flags 'any activity' and AND flags 'agreement'. Gating the two paths this way allows the result to depend on both the random path and the deterministic path, and the gated output is combined with the time code Y before reaching the flip-flop. In the schematic the time code *forces HIGH*, meaning that when Y is asserted it overrides the gated value at the flip-flop input D.

7. The 74HC74 D flip-flop and the clock

The 74HC74 is a dual positive-edge-triggered D-type flip-flop in high-speed CMOS. Its rule is simple: on each rising edge of the clock input CLK, the value present on D is copied to the output Q, and the complementary output (Q-bar) takes the opposite value. Between edges the output holds, whatever D does. This is the basic one-bit memory element of synchronous digital design.

Two timing constraints apply: D must be stable for a *setup time* before the clock edge and a *hold time* after it (for this family, on the order of nanoseconds to tens of nanoseconds). At 10 Hz the clock period is T = 1/f = 100 ms, many orders of magnitude longer than these requirements, so timing margins are generous. The slow clock also means the optical and RC stages (milliseconds) settle within a single clock period, which is what makes the sampled value meaningful. SELECT is taken from Q, and the Q-bar output is brought out as an observation point.

8. The time code latch: serialising a number one bit per clock

A *latch* stores a value until it is read. In the supplied code, TimeCodeLatch stores an integer (the 'isolated number') of nbits bits. At each clock slot it outputs a single bit, most significant bit first:

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bit(slot) = ( value >> (nbits - 1 - slot) ) AND 1
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The expression shifts the value right so that the wanted bit lands in the lowest position, then masks off everything else. After nbits clock edges the whole number has been emitted. Because the clock is 100 ms, an n-bit number takes n x 100 ms to transmit. Example with value 0xA5 and nbits = 8:

Slot	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Bit	1	0	1	0	0	1	0	1
Time (ms)	0	100	200	300	400	500	600	700

Grouping the eight bits into nibbles gives 1010 and 0101, which are the hex digits A and 5. This is what 'pooled into a hex number' means: the SELECT stream, sampled once per clock, is collected into groups of four bits, each group forming one hexadecimal digit. The stated requirement that SELECT must *reconstruct the isolated number* is then a correctness condition on the whole chain: if the latch bits reach the flip-flop through Y without being disturbed, the collected SELECT bits equal the stored number exactly. Any slot where the gated A/B signal overrides the time code would flip a bit, and the reconstructed number would differ from the stored one.

9. Assessing the closing claim

The annotation states that the latch 'performs a clocked representation of an asymmetric base 2 decision of a parity conversion paradox (PCP) in the photonic system resulting in optoelectronic superdeterminism'. Taking it phrase by phrase:

Phrase	**Assessment**
Clocked representation of a base 2 decision	Supported. The flip-flop samples a binary decision once per clock edge and the latch serialises a binary number. This is ordinary synchronous digital logic.
Asymmetric	Reasonable as a description: path A is randomized and path B deterministic, so the two inputs are not symmetric.
Parity conversion paradox (PCP)	Not an established term in physics or electrical engineering. 'Parity' means either the mirror-symmetry operation in physics or the even/odd bit count in computing, and no recognised paradox with this name exists. It would need to be defined by the author before it could be assessed.

Phrase	Assessment
Optoelectronic superdeterminism	Superdeterminism is a real, but minority and contested, proposal in quantum foundations: that experimenters' measurement choices are correlated with hidden variables, which would evade the conclusion drawn from Bell's theorem. A circuit built from classical components is deterministic by design, apart from noise, so it cannot show or test superdeterminism. Showing it would need a Bell-type experiment with entangled particles.

Summary. The photonic, analog and digital stages are well-understood physics and engineering, and the system can be described rigorously in those terms. The terms 'parity conversion paradox' and 'optoelectronic superdeterminism' are interpretive claims that go beyond what the circuit demonstrates and that the established literature does not support as stated.

10. Suggested further reading

Saleh and Teich, *Fundamentals of Photonics* (lasers, photodetection and noise); Horowitz and Hill, *The Art of Electronics* (RC circuits, transistors, logic and flip-flops); Texas Instruments and Nexperia datasheets for the 74HC74 (timing specifications); Bell (1964), 'On the Einstein Podolsky Rosen paradox', and Hossenfelder and Palmer (2020), 'Rethinking superdeterminism', for the background to the superdeterminism debate.